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## Session 4 (2h) — Parametric Design & Tolerancing

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### Core concepts (1h)

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- **파라메트릭 설계 (parametric design)** — instead of hard-coding numbers into every dimension, you define **user parameters** (named variables, e.g. `wall_thickness = 2mm`) and reference them in sketches/features. Change the parameter once, and every feature that uses it updates automatically.
- **Why this matters for prototyping** — internship work usually means multiple print-and-fit iterations. Parametric models let you tweak one number (e.g., “hole diameter”) instead of redoing five features by hand.
- **치수 설계 (tolerance/fit design)** — 3D-printed parts are never dimensionally perfect. Standard practice:
  - Holes for screws/pins: add +0.2–0.3mm to the nominal diameter
  - Two parts meant to slide: add ~0.2–0.4mm clearance per side
  - Two parts meant to press-fit: keep them exact or slightly under (0–0.1mm) — printer- dependent, always test first
- **Design table / parameter list** — Fusion’s “Change Parameters” dialog lists every named parameter in one place; on-site teams use this to standardize dimensions across parts.

### Hands-on practice (1h)

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1. Open the enclosure from Session 2–3. Create user parameters: `wall_thickness`, `enclosure_height`, `screw_hole_dia`.
2. Re-edit the shell and extrude features to reference these parameters instead of raw numbers.
3. Add 4 mounting holes sized `screw_hole_dia` (e.g., set to M3 screw diameter + 0.3mm clearance).
4. Change `wall_thickness` from 2mm to 3mm in the parameter list and confirm the whole model updates without re-doing any feature.
5. Note in your file: which dimensions you’d tell a teammate are “tolerance-critical” vs cosmetic.

### Checkpoint

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- ☐ Created and used at least 3 named user parameters
- ☐ Can explain why printed holes need positive tolerance and press-fits need near-zero
- ☐ Verified the model updates correctly when a parameter changes